

Guide: What the Affine Transfer Would Prove

Collatz proof project — September 5, 2026

The result

Lean now verifies that the full Collatz conjecture is equivalent to this single transfer statement:

$$\begin{aligned} z \equiv 2 \pmod{3} \text{ and } z \text{ reaches one} \\ \implies 9z + 2 \text{ reaches one.} \end{aligned}$$

Equivalently, reaching one must transfer from $3u + 2$ to $27u + 20$ for every natural u . The implication itself has not been proved. This is an exact reformulation of the goal, not a completed proof.

Why this pair appears

Compare an odd seed n with the smaller seed $n - 1$. The previous classical odd-run result gives two possibilities. In one case they merge. In the other, their later values are $9z + 2$ and z . After excluding the short cases that already descend, the smaller endpoint z is always two modulo three.

If every smaller positive seed reaches one, then $n - 1$ and its later value z do too. The proposed transfer would make $9z + 2$ reach one, and therefore make n reach one. Lean checks this argument by strong induction, including the even seeds and the short odd-run cases.

Boundedness is a different version

Replacing “reaches one” with “has a bounded orbit” gives an equivalent reformulation of nondivergence. In fact, if any unbounded orbit exists, Lean proves that there must be a bounded $z \equiv 2 \pmod{3}$ whose partner $9z + 2$ is unbounded.

Boundedness alone allows nontrivial cycles. The reaching-one version addresses them as well, so it matches the full original goal. We should not treat the two transfer statements as interchangeable without proof.

What this changes in the research plan

The affine transfer cannot be assumed as a routine stability property. Even its restriction to one residue class contains the full unresolved claim. The useful output is a precise target and a proof that it would suffice. A future argument must explain why reaching one transfers between these partners for every allowed input.

Finite examples cannot establish that statement: both members may converge in every tested pair while the universal implication remains unproved. The new Lean proofs use no experimental premise and do not assert any progress on establishing the transfer itself.

Where to check the proofs

The main theorem is `collatz_iff_restricted_affine_transfer` in `lean/Collatz/AffineConvergence.lean`. The boundedness equivalences and failure-pair construction are in `lean/Collatz/AffineBoundedness.lean`. Build these modules from `lean/`. The repository theorem audit checks their statements and axiom dependencies. The technical paper gives the proof structure and relates it to the classical odd-run identities.